2023

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-711

(Elective I : Internet of Things)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1, 2** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5
   (a) What is the difference between Industry 4.0 and Industry 5.0?
   (b) Compare the relative advantages and disadvantages of COAP and MQTT.
   (c) What are the advantages of using R-Pi over arduino?
   (d) What are the advantages of using MongoDB?
   (e) Mention four advantages of using Apache Spark.
   (f) What is edge computing?

2. Answer ***any five*** questions : 4×5
   (a) Explain the term 'Carbon Footprint'. Why it is important to consider after the onset of ChatGPT?
   (b) What are the different types of Analytics?
   (c) Briefly explain the role of Actuators in IoT.
   (d) Compare Zigbee and Bluetooth.
   (e) Compare XMPP vs Web socket.
   (f) What are the differences between data analysis and data analytics?

3. (a) What is data protocol and network protocol?
   (b) Briefly describe the advantages and disadvantages of two networks and two data protocols. 2+8

4. Thinking the use case of Agriculture, explain the schematic design of IoT design methodology. 10

5. (a) Define Machine Learning.
   (b) Explain tasks, performance metrics, and experience.
   (c) What is the Role of Machine Learning in AI? 2+6+2

Please Turn Over

6. (a) Describe the 4 pillars of Industry 4.0.

(b) How can LLMs impact job professionals? 8+2

7. (a) What is IIoT?.

(b) What is symbiotic trust?

(c) Give one example where Manufacturing industry has benefitted from IIoT. 2+3+5

8. (a) What are the difference between Paas, Saas, IaaS, DIaaS?

(b) What are the concerns of cloud computing?

(c) Why are different performance metrics required in different test cases? 3+3+4